Rahul Yedida

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Education

North Carolina State University — Raleigh, NC
Ph.D. Computer Science — *Aug 2019 - Jul 2024*

- Advisor: *Dr. Tim Menzies*
- Dissertation: *Guidelines for the Application of Neural Technologies in Software Analytics (or: How to Do More with Less in SE)*

PES University — Bangalore, India
B.E. Computer Science – Advisor: Dr. Snehanshu Saha — *Aug 2015 - May 2019*

Employment

LexisNexis Legal & Professional — Raleigh, NC
Senior Data Scientist I — *May 2024 - Present*

- Reduced customer-facing product runtime by 24.8% and cut time spent at peak memory from 86.4% to 68% of the runtime via lazy module loading, LLM call profiling, and parallelization.
- As part of a 3-person team, improved complaint drafting usefulness from 40% to 86% and motion drafting from 59.2% to 76%.
- Led the migration from RAG to an agentic legal drafting system as tech lead, and own its continued development.
- Developed a fast (median 12ms) passage filtering approach based on contrastive loss.
- Built team developer tooling: a Rust MITM proxy with request replay, a web UI for visualizing intermediate agent steps, and custom Python lint rules beyond `ruff`'s rule set.
- Mentored an intern on a large-scale, full factorial experiment investigating Pareto-optimal configurations for draft quality and latency.
- **Technology:** Python, Litestar, Google ADK, React, Rust, TypeScript, Tailwind

North Carolina State University — Raleigh, NC
PhD Student — *Aug 2019 - Jul 2024*

- **State-of-the-art hyper-parameter optimization:** Proposed a novel hyper-parameter optimization method that outperforms prior SOTA and is 36.4% faster.
- **Better, faster deep learning for SE:** Improved defect prediction by up to 123% (F-1 score), code smell detection by up to 30% (AUC), issue lifetime prediction by up to 76% (accuracy), automated microservice partitioning by up to 285% (modularity) compared to prior SOTA.
- **Semi-supervised learning:** Achieved state-of-the-art results (up to 100% improvement in AUC) on static code warnings analysis using 10% of the labels.
- **Teaching:** TA for 830 students over 5 semesters across CSC 230 (C and Software Tools), CSC 510 (Software Engineering), and CSC 591/791 (Automated Software Engineering)

Amazon — New York, NY / Bellevue, WA
Software Dev Engineer Intern — *May 2023 - Aug 2023*

- Implemented and shipped profile locks for Prime Video on Echo Show devices.
- **Technology:** React Native, TypeScript

Software Dev Engineer Intern — *May 2022 - Jul 2022*

- Developed a full-stack system to publish announcements in scorecards used by delivery service partners.
- **Technology:** React/Redux, TypeScript, Redux Saga, DynamoDB, Java Spring

Recent Publications

See full list on Google Scholar.

1. **Yedida, R.**, & Menzies, T. (2025). Is Hyper-Parameter Optimization Different for Software Analytics? *IEEE Transactions on Software Engineering, 51(6).*

2. Baldassarre, M. T., Ernst, N., Hermann, B., Menzies, T., & **Yedida, R.** (2023). (Re)use of Research Results (is Rampant). *Communications of the ACM, 66(2), 75-81.*

3. **Yedida, R.**, Kang, H. J., Tu, K., Lo, D., & Menzies, T. (2023). How to Find Actionable Static Analysis Warnings: A Case Study with FindBugs. *IEEE Transactions on Software Engineering, 49(4), 2856-2872.*

4. **Yedida, R.**, Krishna, R., Kalia, A., Menzies, T., Xiao, J., & Vukovic, M. (2023). An Expert System for Redesigning Software for Cloud Applications. *Expert Systems with Applications, 219, 119673.*

5. **Yedida, R.**, Menzies, T. (2022). How to Improve Deep Learning for Software Analytics (a case study with code smell detection). In *2022 IEEE/ACM 19th International Conference on Mining Software Repositories (MSR).*

6. **Yedida, R.**, & Menzies, T. (2021). On the Value of Oversampling for Deep Learning in Software Defect Prediction. *IEEE Transactions on Software Engineering, 48(8), 3103-3116.*

7. Agrawal, A., Yang, X., Agrawal, R., **Yedida, R.**, Shen, X., & Menzies, T. (2021). Simpler Hyperparameter Optimization for Software Analytics: Why, How, When?. *IEEE Transactions on Software Engineering, 48(8), 2939-2954.*

8. Yang, X., Chen, J., **Yedida, R.**, Yu, Z., & Menzies, T. (2021). Learning to recognize actionable static code warnings (is intrinsically easy). *Empirical Software Engineering, 26(3), 1-24.*

9. **Yedida, R.**, Krishna, R., Kalia, A., Menzies, T., Xiao, J., & Vukovic, M. (2021). Lessons learned from hyper-parameter tuning for microservice candidate identification. *Proceedings of the thirty-sixth IEEE/ACM International Conference on Automated Software Engineering (ASE).*

10. **Yedida, R.**, & Saha, S. (2021). Beginning with Machine Learning: A Comprehensive Primer. *The European Physical Journal Special Topics*, 230(10), 2363-2444.

11. Saha, S., Nagaraj, N., Mathur, A., **Yedida, R.**, & Sneha, H. R. (2020). Evolution of novel activation functions in neural network training for astronomy data: habitability classification of exoplanets. *The European Physical Journal Special Topics, 229(16), 2629-2738.*

12. **Yedida, R.**, Saha, S., & Prashanth, T. (2020). LipschitzLR: Using theoretically computed adaptive learning rates for fast convergence. *Applied Intelligence, 1-19.*

13. Sridhar, S., Saha, S., Shaikh, A., **Yedida, R.**, & Saha, S. (2020, July). Parsimonious Computing: A Minority Training Regime for Effective Prediction in Large Microarray Expression Data Sets. In *2020 International Joint Conference on Neural Networks (IJCNN) (pp. 1-8).*

14. Khaidem, L., **Yedida, R.**, & Theophilus, A. J. (2019, November). Optimizing Inter-nationality of Journals: A Classical Gradient Approach Revisited via Swarm Intelligence. In *International Conference on Modeling, Machine Learning and Astronomy (pp. 3-14). Springer, Singapore.*

Preprints

1. Hearn, T., Evans, W., **Yedida, R.**, Abdelshiheed, M., Rebei, A., & Khazaeli, S. (2025). Automated Legal Brief Drafting via Iterative Retrieval-Augmented Generation.

2. Rashid, P., Pal, S., Samudrala, V. K., **Yedida, R.**, & Gehringer, E. (2025). Enhancing Peer Assessment Reliability: A Probabilistic Approach Integrating Formative and Summative Feedback.

3. Mehendale, S., Challa, A., **Yedida, R.**, Danda, S., Sarkar, S., & Saha, S. (2025). Radon–Nikodym Derivative: Re-imagining Anomaly Detection from a Measure Theoretic Perspective.

4. **Yedida, R.**, & Saha, S. (2024). Strong convexity-guided hyper-parameter optimization for flatter losses.

Funding

$5,000, Google Cloud Academic Research Grant, Feb 2022

Service to Profession

Contributing Member, Python Software Foundation (Mar 2021 - Present)

Guest Editor, IEEE Software SI: The Impact of AI on Productivity and Code 2025; ASEj SI: Replications and Negative Results (RENE) 2025; EMSE SI: Replications and Negative Results (RENE) 2025

Workshop Co-Chair, Intelligent Software Engineering (ASE '25, ICST '26); Replications and Negative Results (ASE '24)

PC Member, AAAI 2025-2027; FSE 2026-2027; ICSE 2026-2027; EDM 2026; ASE Industry Track 2026; AI Foundation Models and Software Engineering (FORGE) 2024; ICSME Artifact Evaluation Track, 2021-2023; ASE Artifact Evaluation Track, 2022; International Conference on Modeling, Machine Learning, and Astronomy (MMLA), 2019

Reviewer, TOSEM 2026; TSE 2025-2026; NeurIPS 2023, 2025; EMSE 2021, 2025; ASEj 2025; ICML 2024-2025; ICLR 2024-2025; NCAA 2023-2025; TMLR 2024; Neural Processing Letters 2023-2024; Artificial Intelligence Review 2023; J. Big Data, 2023; ASE 2023; IEEE SSCI 2020

Honors

Distinguished Reviewer, ICSE 2026 Mar 2026

Google Developer Expert - ML and GCP Mar 2025 - Present

A global network of experts recognized by Google for deep technical expertise, community leadership, and mentorship.

Google Cloud Champion Innovator - Cloud AI/ML Oct 2022 - Mar 2025

A Googler-nominated global network of 500+ technical experts in Google Cloud; merged into the Google Developer Experts program in 2025.

Google Cloud Research Innovator Feb 2022

Research Innovators are a global community of researchers driving scientific breakthroughs with Google Cloud.

Invited Talks

Feb 2024, "*Improving deep learning performance using theoretical ML*" at BITS Pilani, KK Birla Goa Campus, India

Skills

Languages: Rust, Python, TypeScript, Java, C++, Gleam

ML/AI: Keras, PyTorch, Hyper-parameter optimization

Frameworks: React, Flask, Litestar, FastAPI

Databases: LanceDB, MySQL, MongoDB, DynamoDB

Cloud: Google Cloud (Compute Engine, Cloud Storage, Vertex AI), AWS (EC2, S3, DynamoDB)